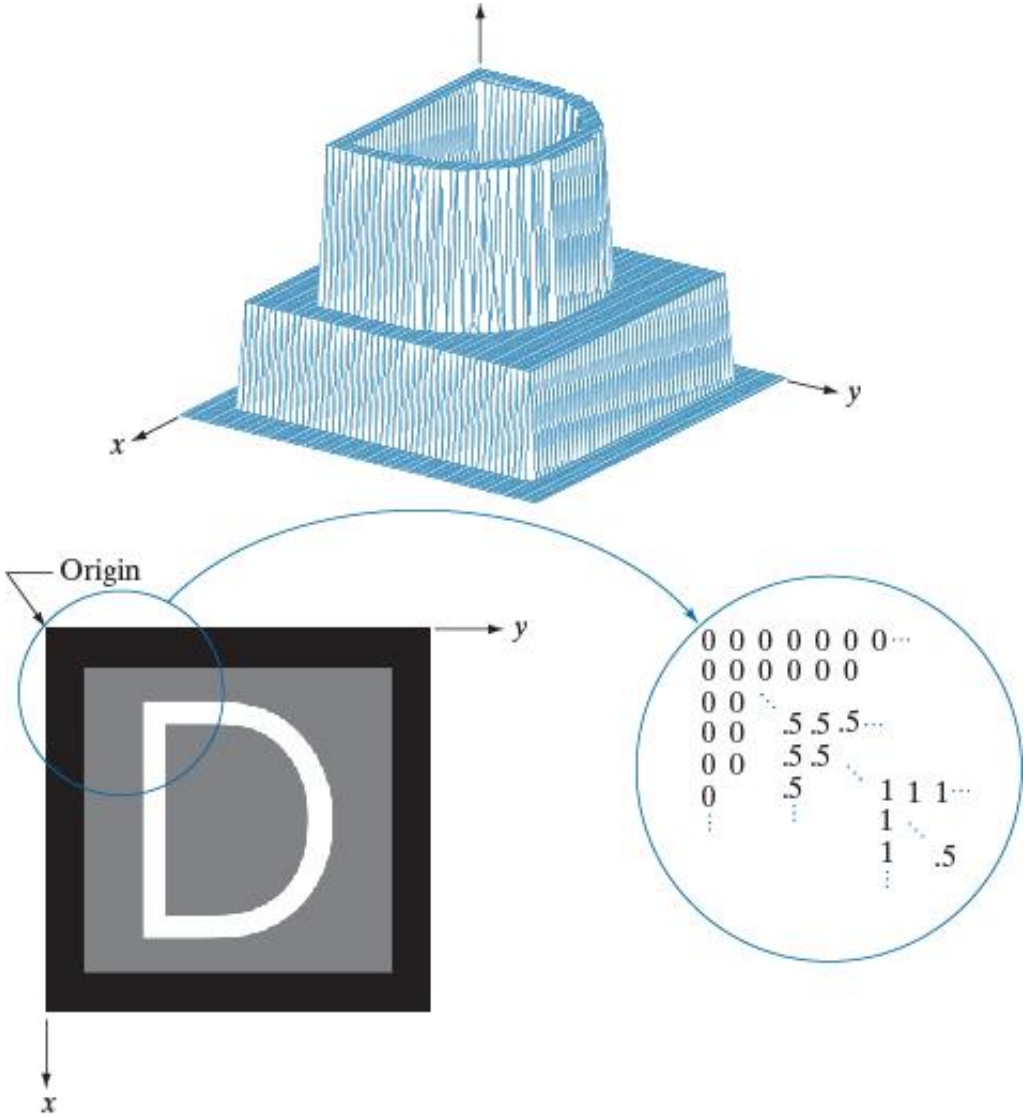
Lecture - 3

Chapter 2

DIGITAL IMAGE FUNDAMENTALS

Digital Image Representation



a
b c

FIGURE 2.18
(a) Image plotted as a surface.
(b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)

Digital Image Representation

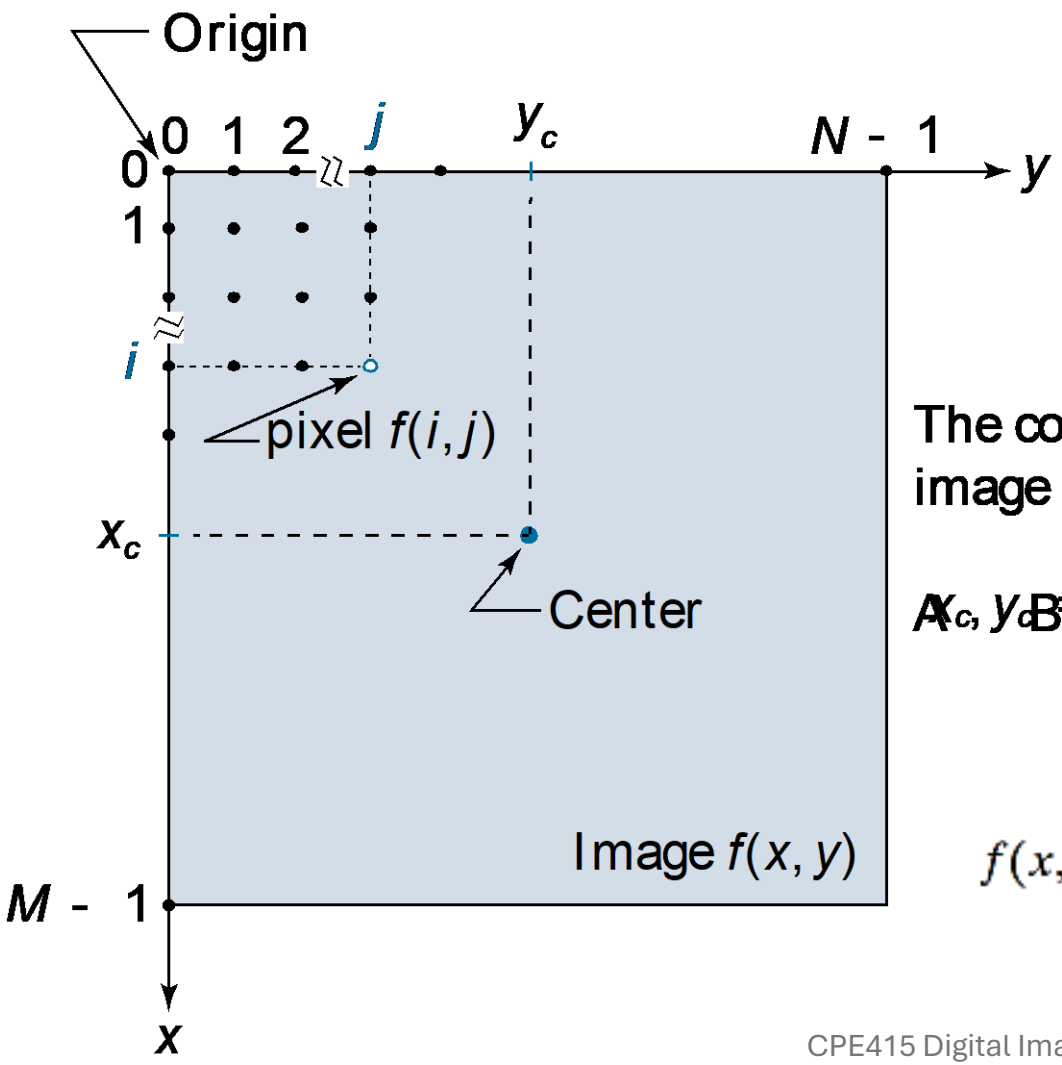


FIGURE 2.19 Coordinate convention used to represent digital images. Because coordinate values are integers, there is a one-to-one correspondence between x and y and the rows (r) and columns (c) of a matrix.

The coordinates of the image center are

$$(x_c, y_c) = (\text{floor}(M/2), \text{floor}(N/2))$$

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

Dynamic Range

- $L = 2^k$ gray levels, gray scales [0,…,L-1]
- The dynamic range of an imaging system is the range between minimum and maximum detectable intensity in the scene.
  - Saturation: maximum measurable intensity (beyond which all intensities are clipped)
  - Noise: minimum detectable intensity (noise)

    If the dynamic range of an image spans a significant portion of the gray scale → high contrast
  - Otherwise, low dynamic range results in a washed out gray look.

Noise and Saturation

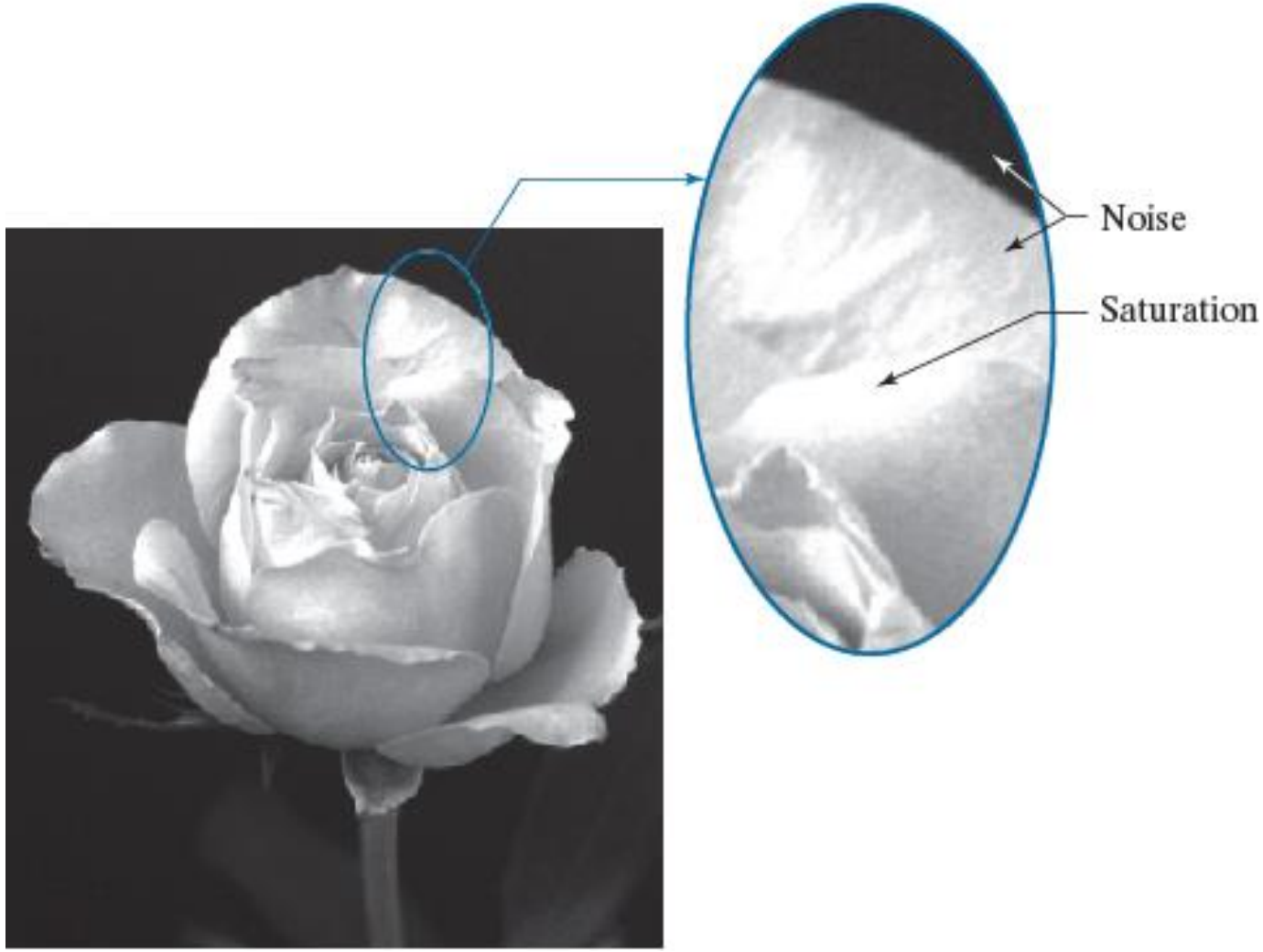


FIGURE 2.20 An image exhibiting saturation and noise. Saturation is the highest value beyond which all intensity values are clipped (note how the entire saturated area has a high, constant intensity level). Visible noise in this case appears as a grainy texture pattern. The dark background is noisier, but the noise is difficult to see.

Image Storage

bits to store the image = N x M x k

gray level = L = 2^k

TABLE 2.1

Number of storage bits for various values of N and k.

N/k	1 (L = 2)	2 (L = 4)	3 (L = 8)	4 (L = 16)	5 (L = 32)	6 (L = 64)	7 (L = 128)	8 (L = 256)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Image Storage

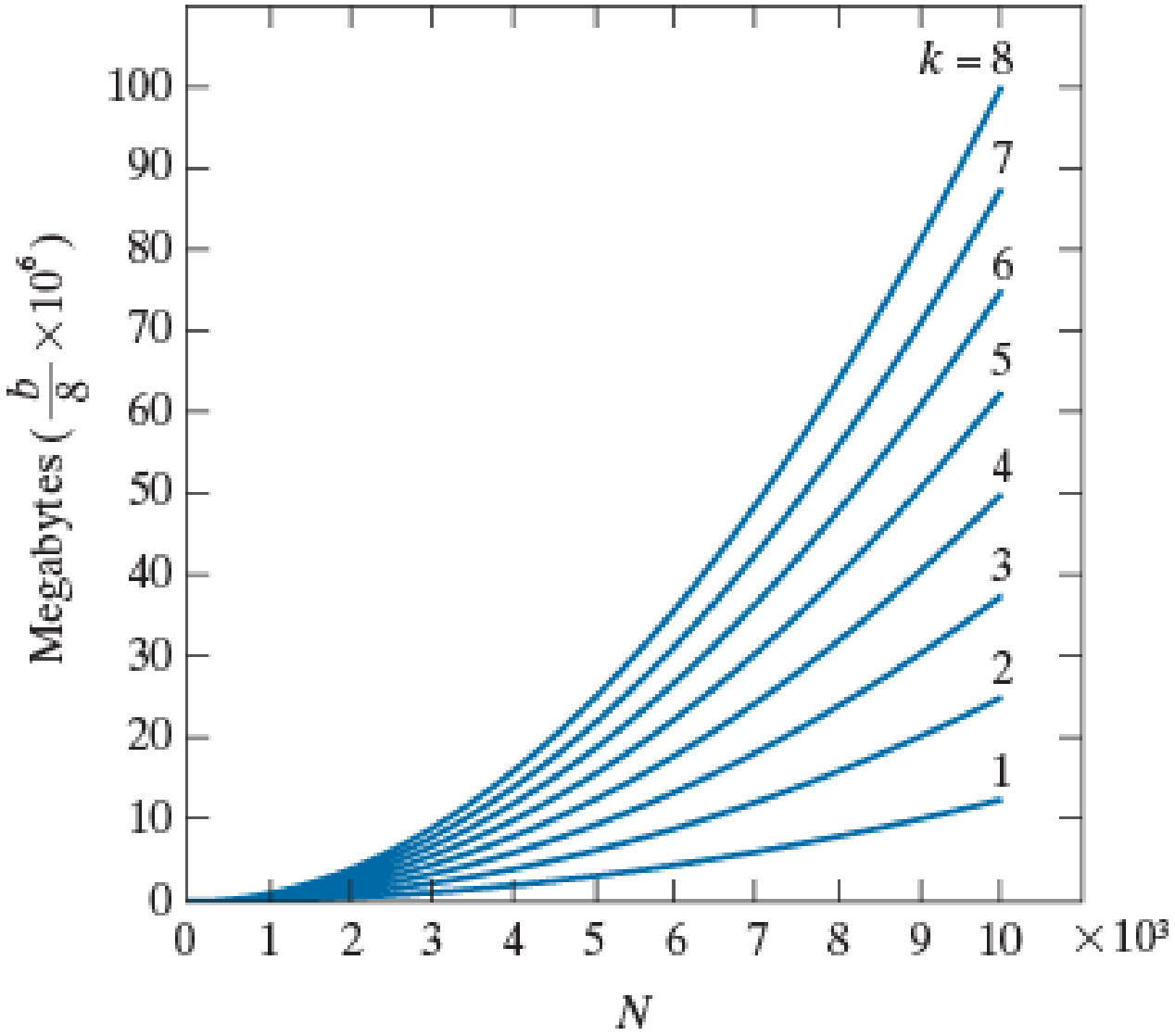


FIGURE 2.21 Number of megabytes required to store images for various values of N and k.

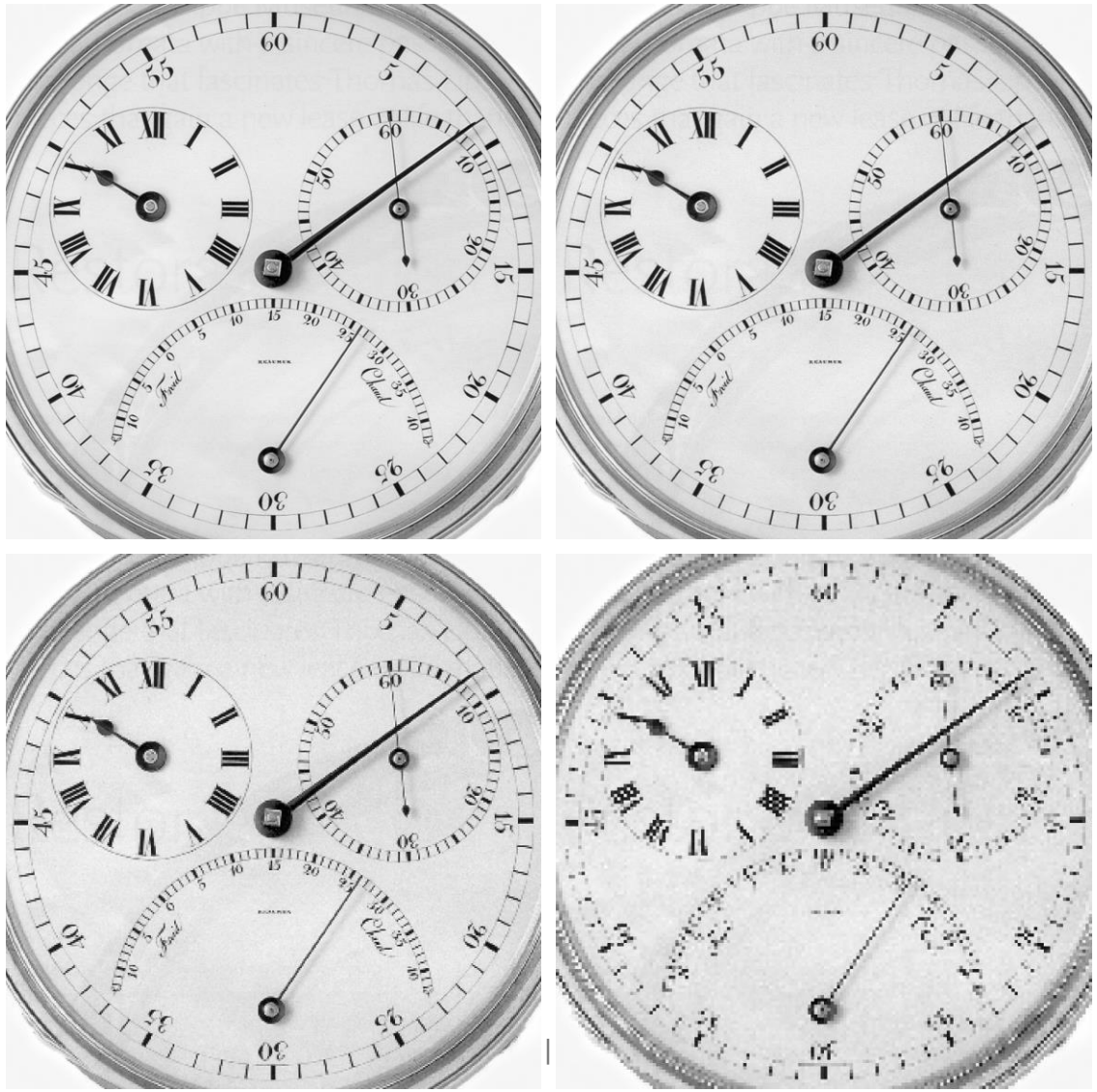
Spatial and Intensity Resolution

- **Spatial Resolution:** Spatial resolution is the smallest detectable detail in an image. (typicaly dpi - dots per inch)
  - dpi – resolution of an image printed on paper
  - ppi – resolution of an image displayed on screen (dpi also used)

- **Grey level Resolution:** Gray-level resolution similarly refers to the smallest detectable change in gray level.

Resolution	Size (Width × Height)
SD	720 × 576
HD (720p)	1280 × 720
Full HD (1080p)	1920 × 1080
Ultra HD (4K)	3840 × 2160
UHD-2 (8K)	7680 × 4320

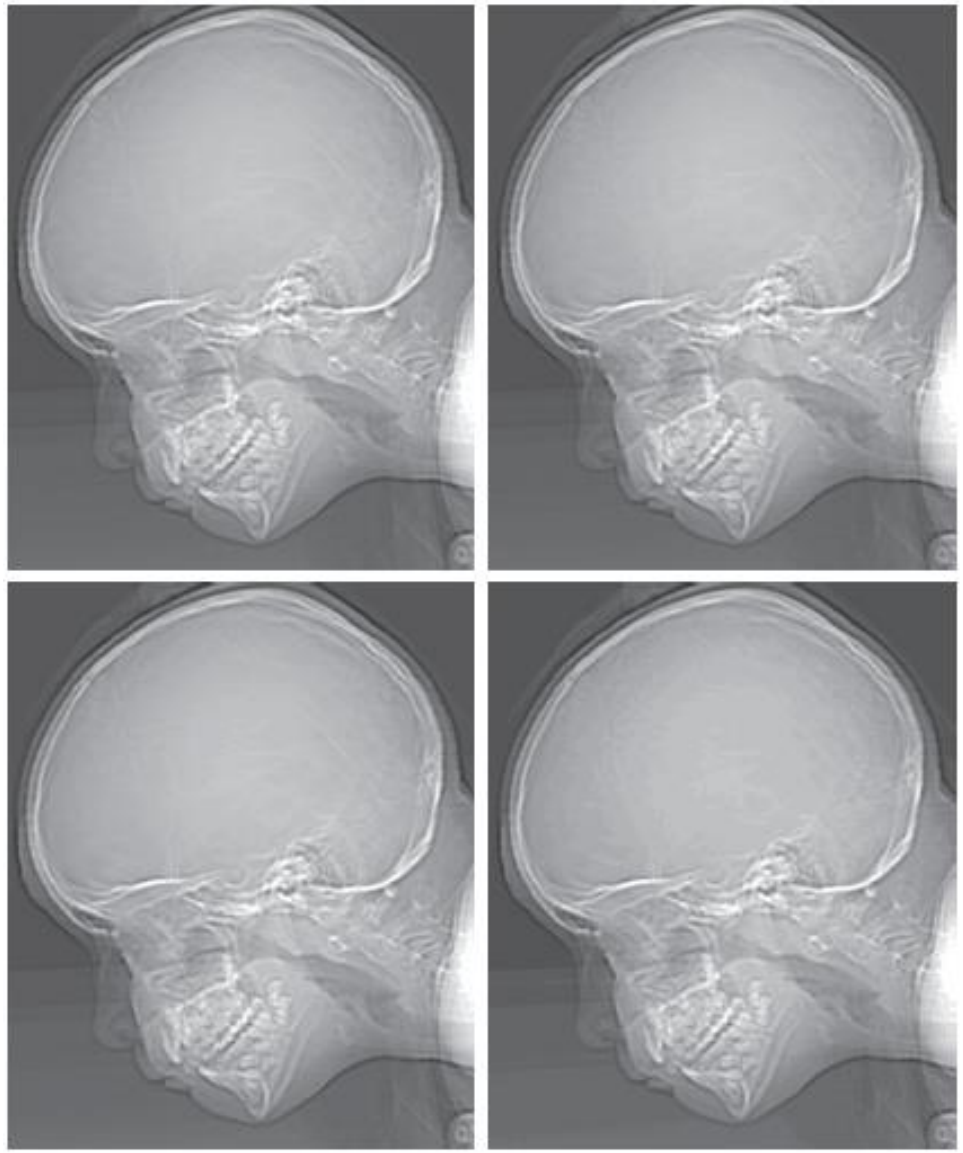
Spatial Resolution



a b
c d

FIGURE 2.23 Effects of reducing spatial resolution. The images shown are at: (a) 930 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi.

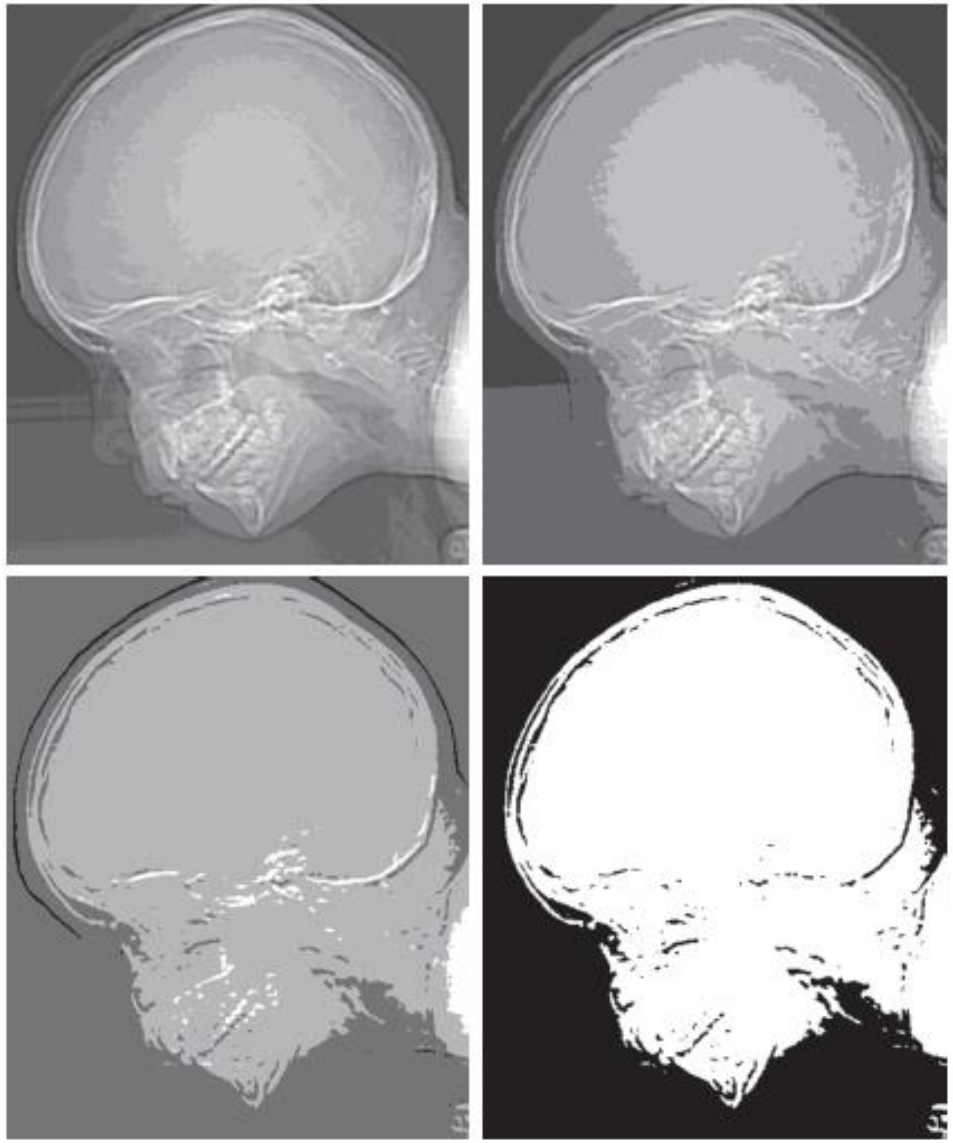
Intensity Resolution



a b
c d

FIGURE 2.24 (a) 774 × 640, 256-level image. (b)-(d) Image displayed in 128, 64, and 32 intensity levels, while keeping the spatial resolution constant. (Original image courtesy of the Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)

Intensity Resolution



e f
g h

FIGURE 2.24 (*Continued*) (e)-(h) Image displayed in 16, 8, 4, and 2 intensity levels.

Spatial and Intensity Resolution (isopreference curves)

a b c

FIGURE 2.25 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

Spatial and Intensity Resolution (isopreference curves)

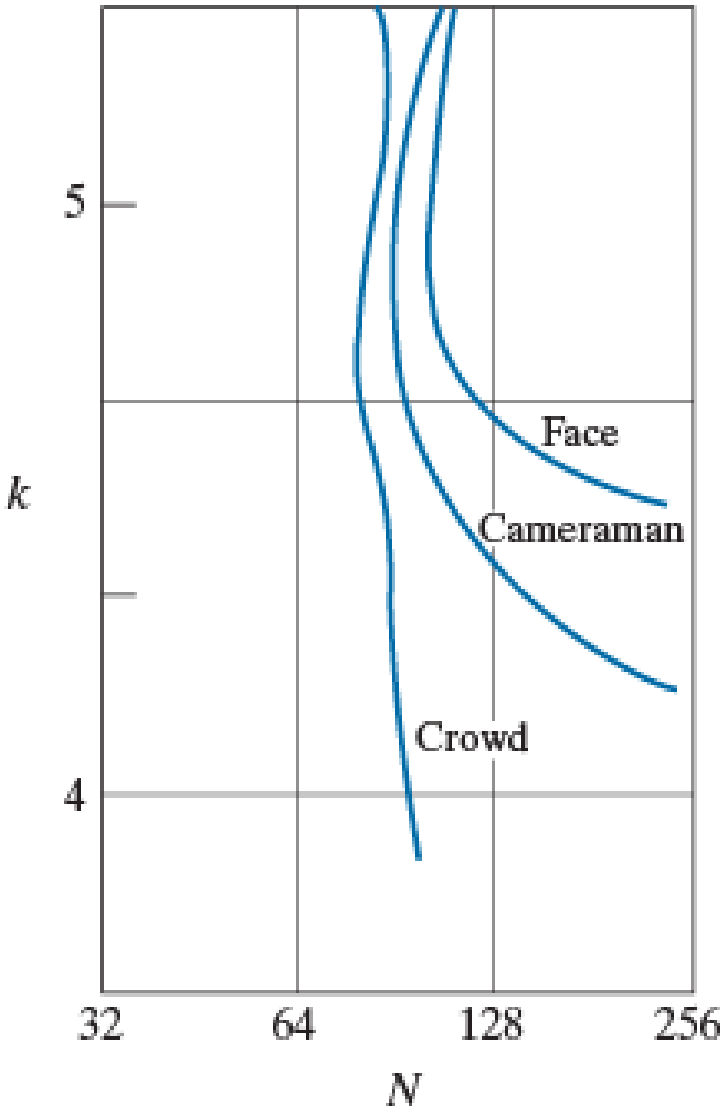


FIGURE 2.26 Representative isopreference curves for the three types of images in Fig. 2.25.

Image Interpolation

Zooming: Oversampling

Creation of new pixel locations

The assignment of gray levels to those new locations

Nearest Neighbor Interpolation

Special case : Pixel Replication, if the size of new image is integer multiple of the original image size

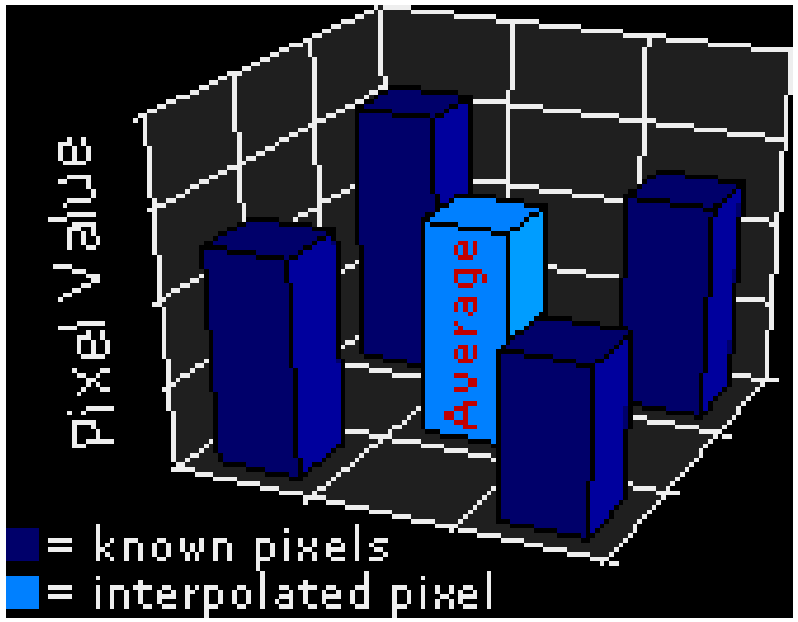
NN is fast but produces checker-board effect

Bilinear Interpolation: uses four nearest neighbors of a point

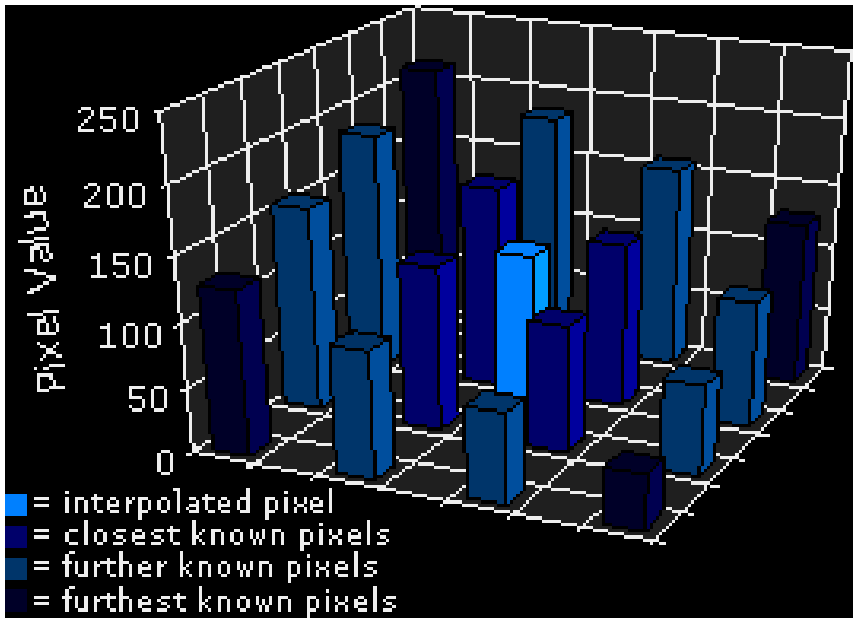
Bi-cubic Interpolation: uses sixteen nearest neighbors of a point

Shrinking: Under sampling

Image Interpolation

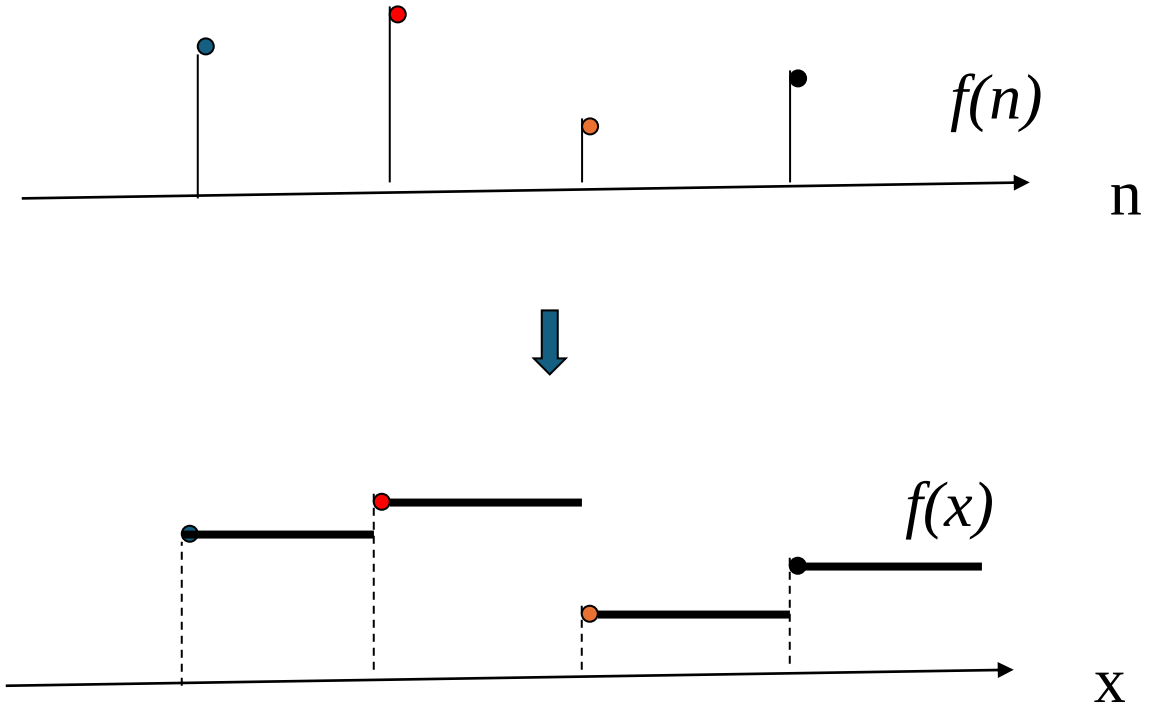


Bilinear interpolation

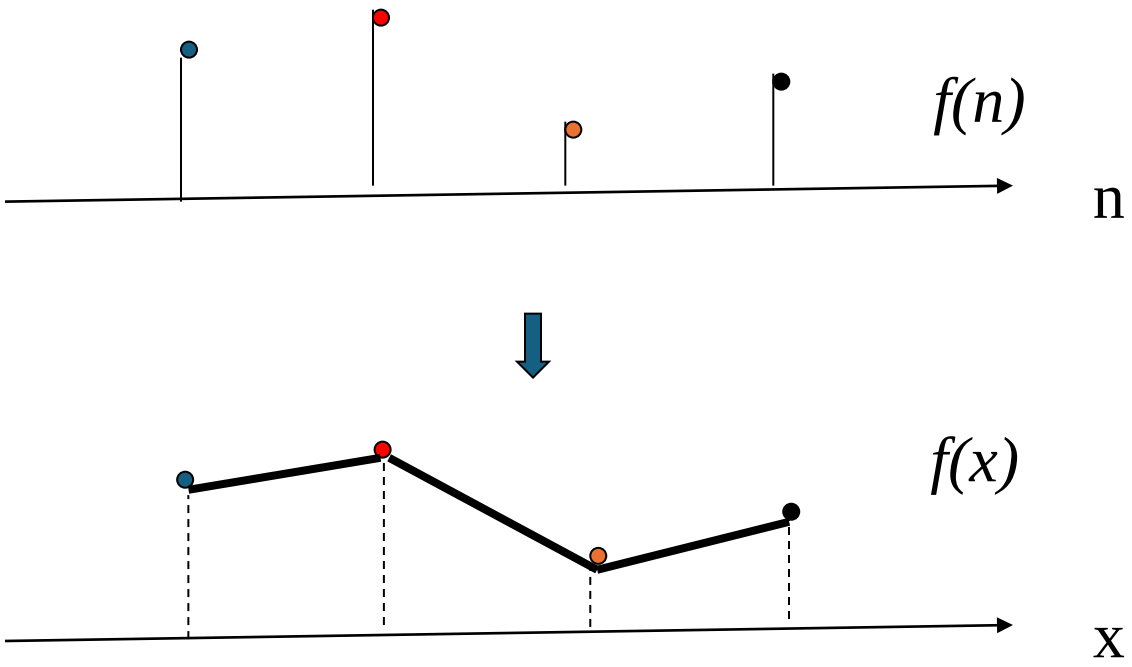


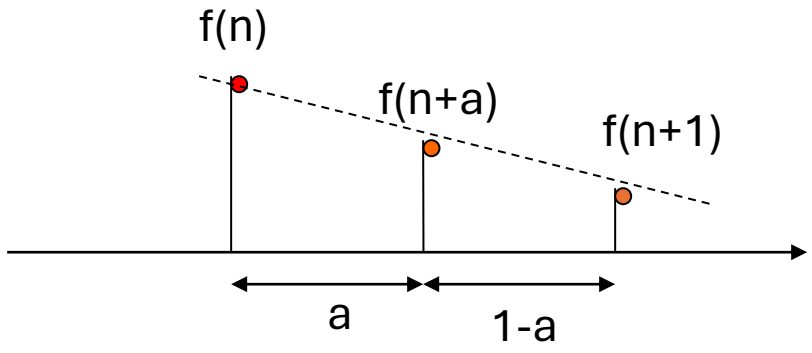
Bicubic interpolation

Nearest Neighbor (Zero Order hold) Interpolation



Linear (1st Order hold) Interpolation





$$f(n+a)=(1-a)\times f(n)+a\times f(n+1),\ 0<a<1$$

Note: when a=0.5, we simply have the average of two

Linear (1st Order hold) Interpolation - Example

f(n)=[0,120,180,120,0]

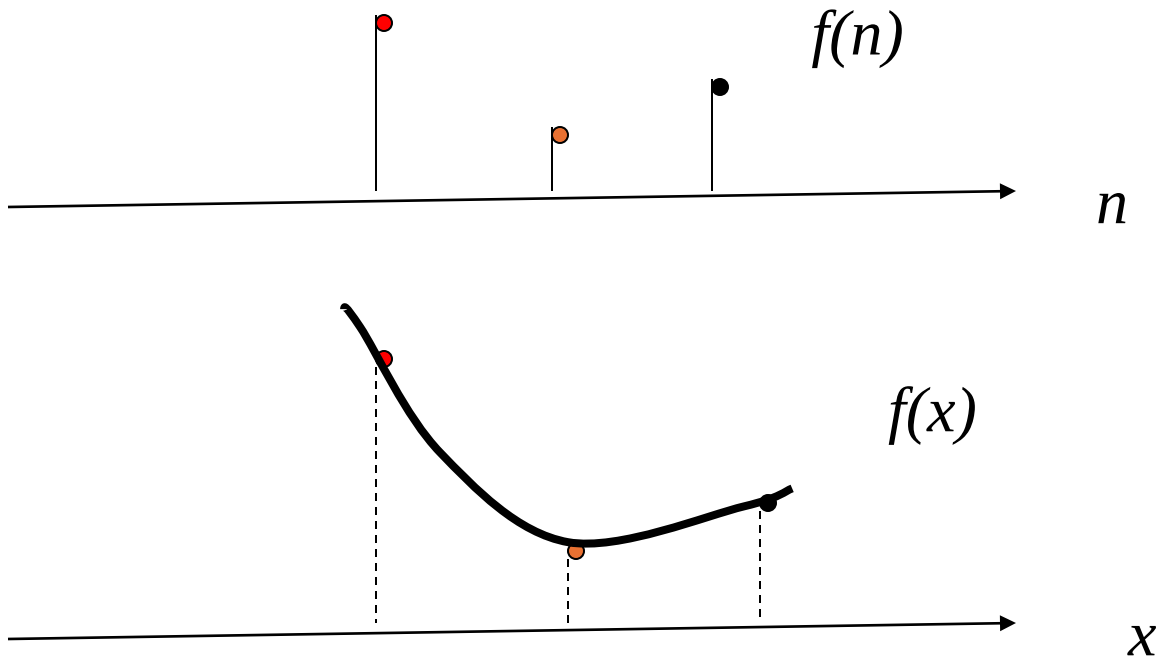
↓ Interpolate at 1/2-pixel

f(x)=[0,60,120,150,180,150,120,60,0], x=n/2

↓ Interpolate at 1/3-pixel

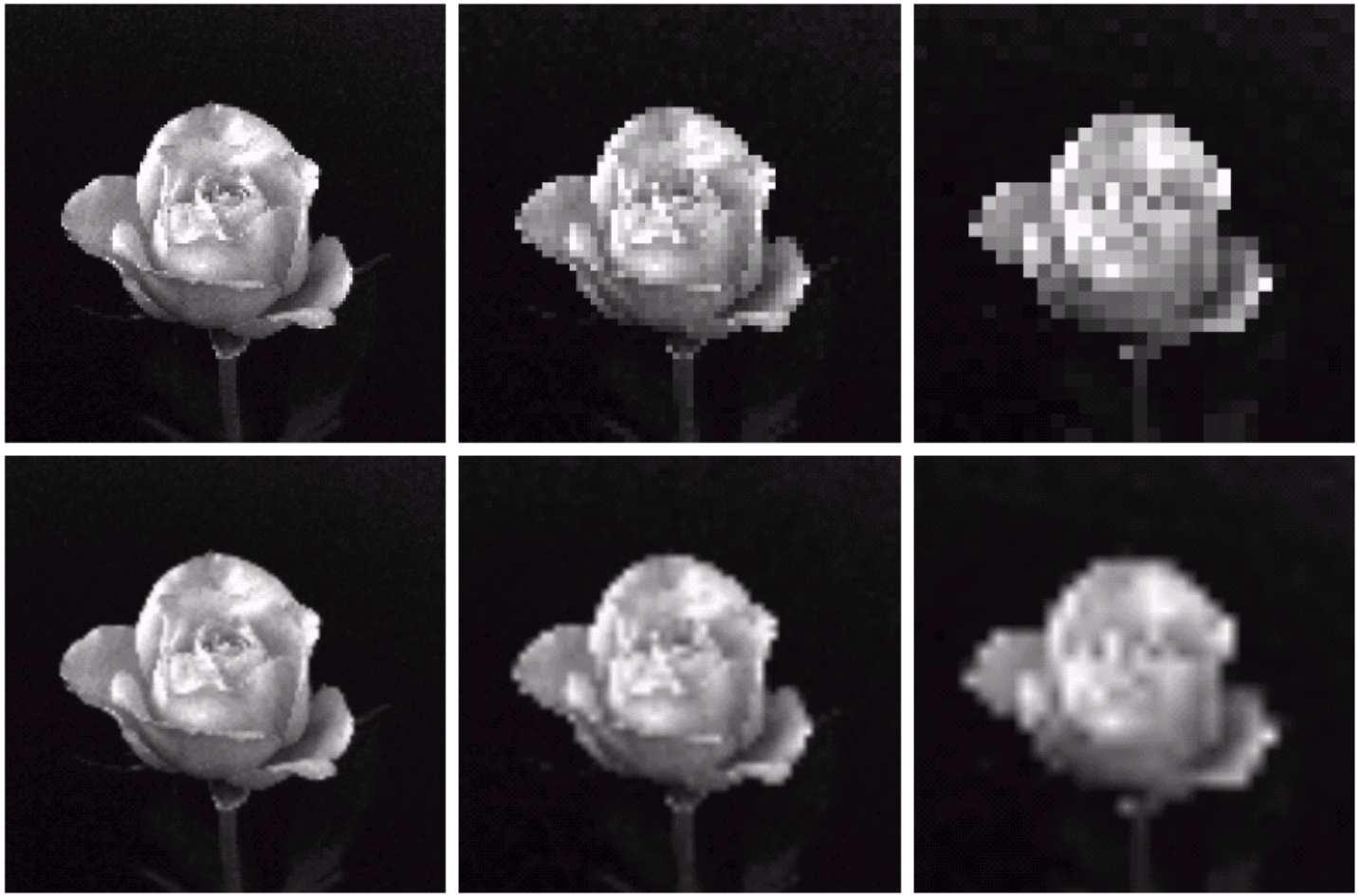
f(x)=[0,20,40,60,80,100,120,130,140,150,160,170,180,…], x=n/6

Cubic Interpolation



Cubic spline fitting

Image Interpolation



a b c
d e f

FIGURE 2.25 Top row: images zoomed from 128 × 128, 64 × 64, and 32 × 32 pixels to 1024 × 1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Image Interpolation

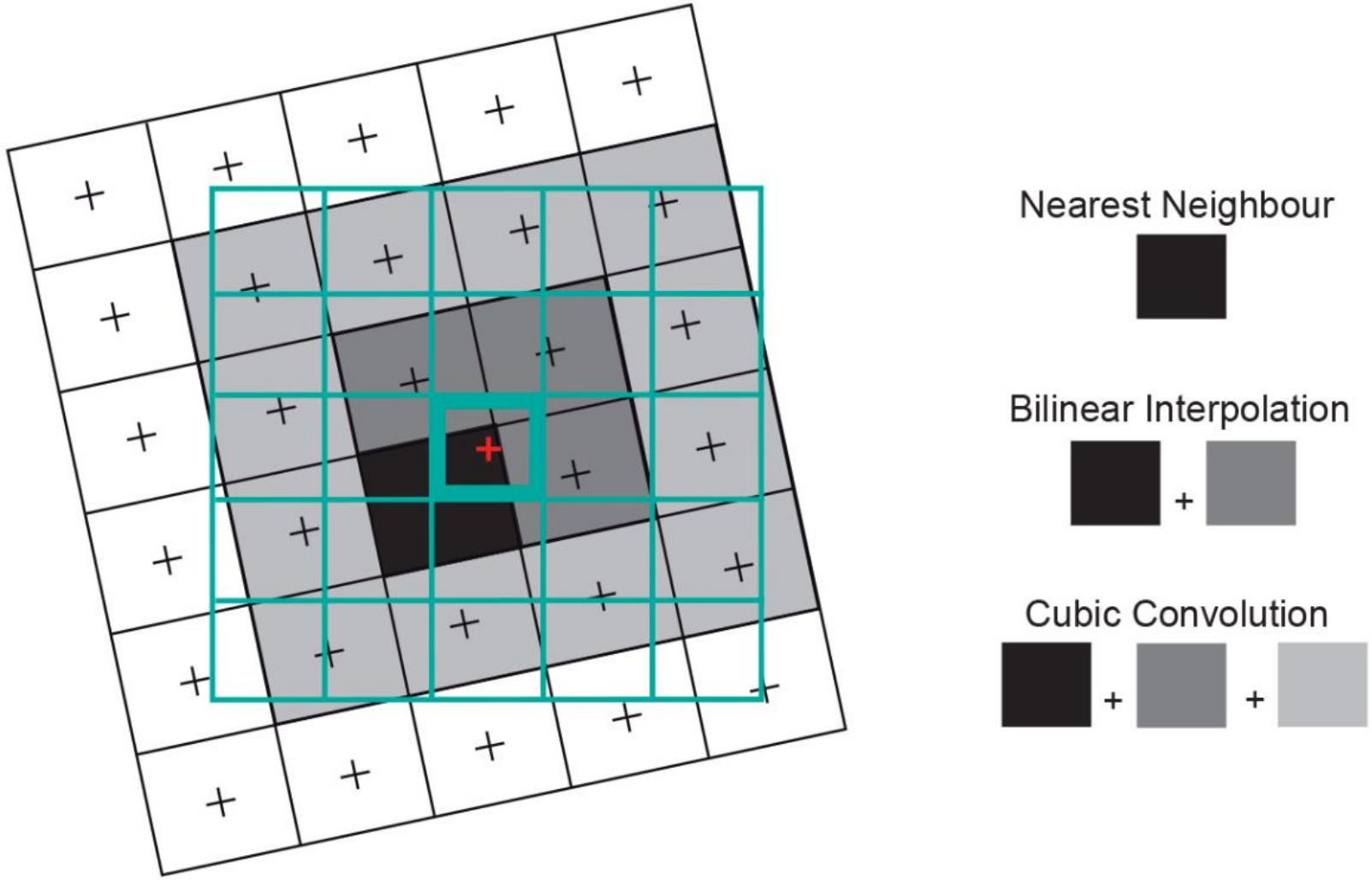
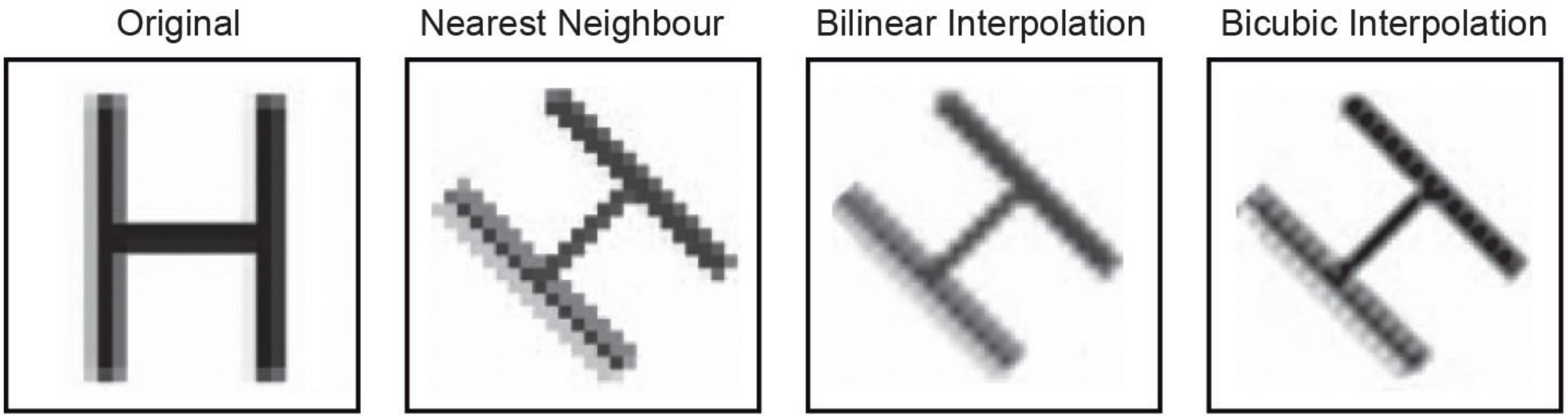



Image Interpolation